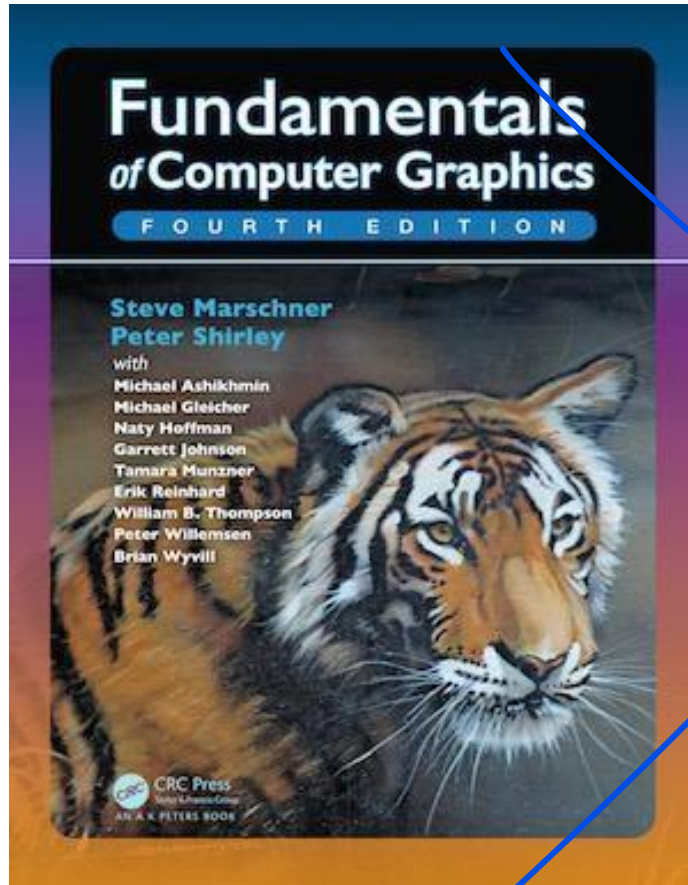


CSE4203: Computer Graphics
Chapter – 1
Introduction

Outline

- What is CG
- CG Areas
- Major Applications
- Graphics API

Credit



CS4620: Introduction to Computer Graphics

Cornell University

Instructor: Steve Marschner

<http://www.cs.cornell.edu/courses/cs4620/2019fa/>

What is CG? (1/1)

- (The term computer graphics describes any use of computers to **create** and **manipulate** images.)
 - Graphics can be 2D or 3D
 - Images can be completely synthetic or can be produced by manipulating photographs.

CG Areas (1/4)

- **Modeling:**
 - deals with the **mathematical specification:**
 - shape and appearance properties in a way that can be stored on the computer.

CG Areas: Metaphor

1



2



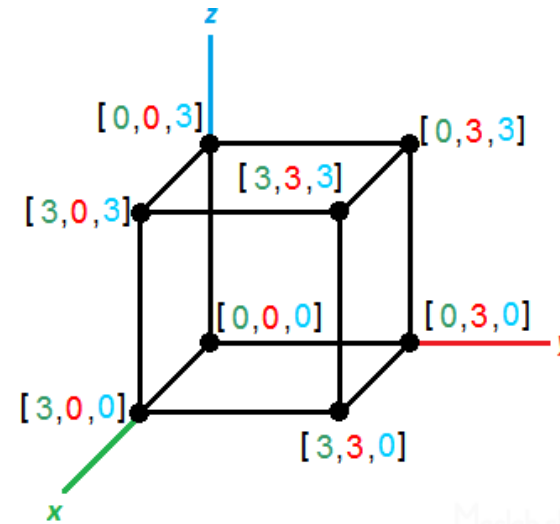
3



Source: <https://youtu.be/6Sv4oXSTams>

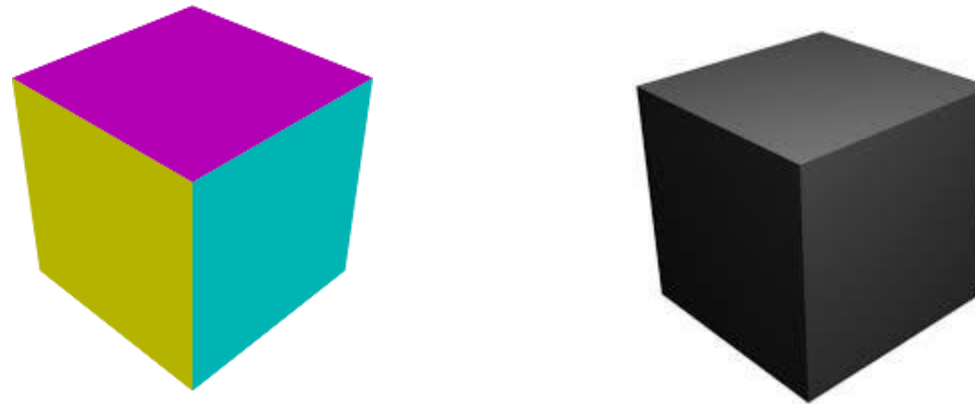
CG Areas (2/4)

- **Modeling:** Example –
 - an object can be described as 3D coordinates:
 $[0, 0, 3], [0, 3, 3], [0, 3, 0], [0, 0, 0],$
 $[3, 0, 3], [3, 3, 3], [3, 3, 0], [3, 0, 0]$
 - connect the points



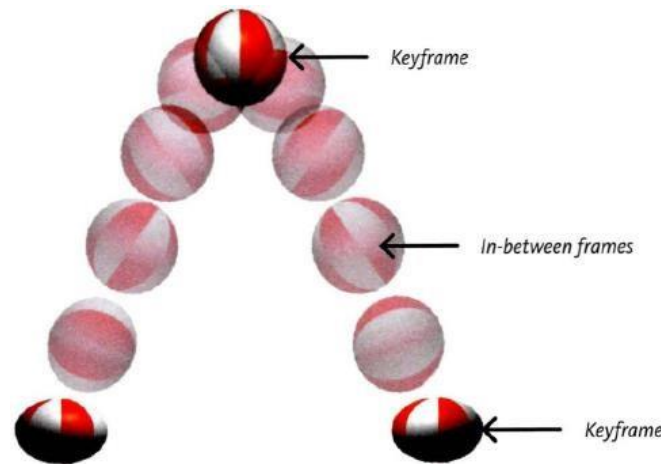
CG Areas (3/4)

- **Rendering:**
 - a term inherited from art
 - deals with the creation of shaded images from 3D computer models.



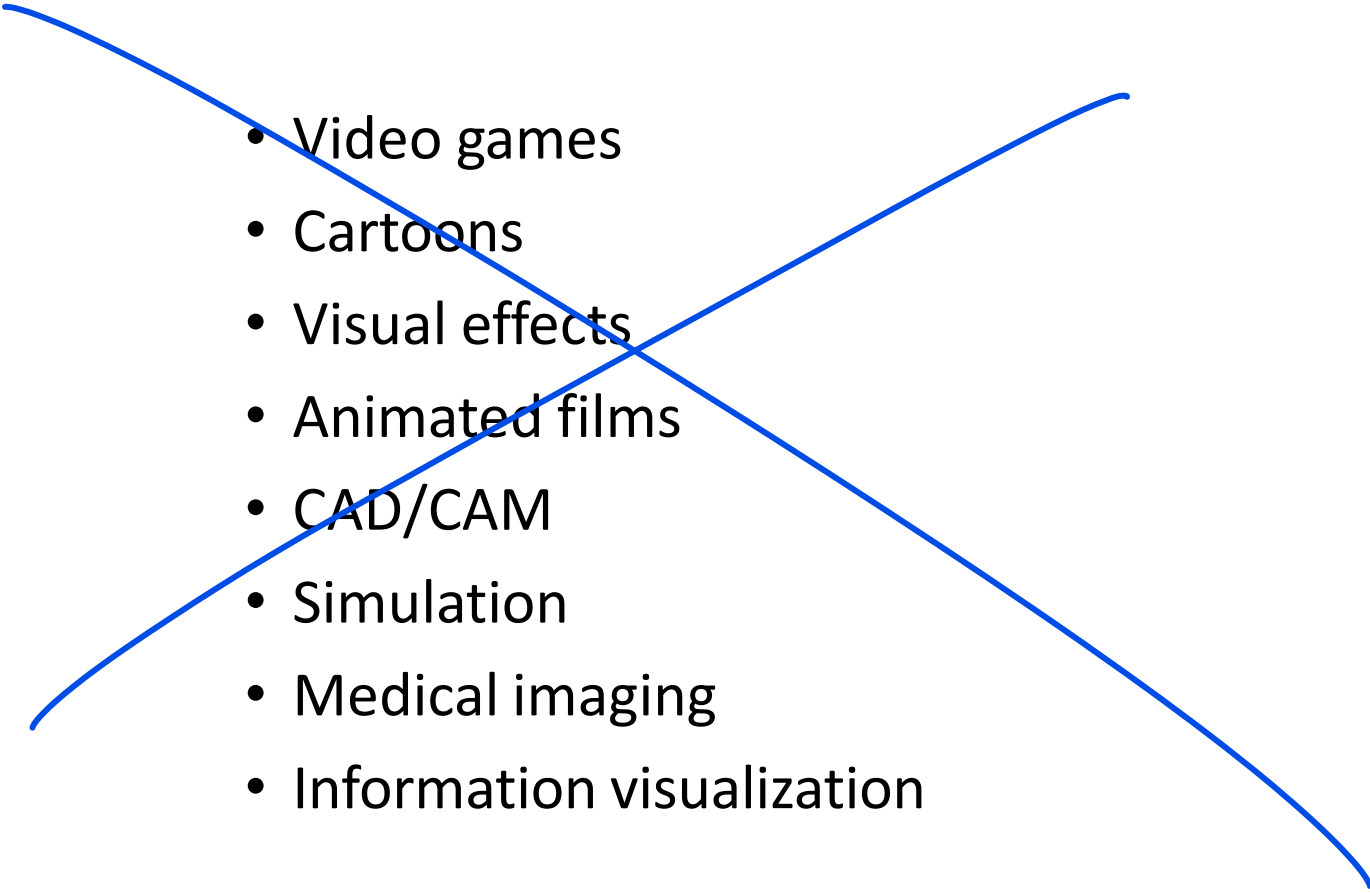
CG Areas (4/4)

- **Animation:**
 - creates an **illusion of motion** through sequences of images.
 - uses modeling and rendering but adds **movement** over time



Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | <http://www.cs.cornell.edu/courses/cs4620/2019fa/>

Major Applications (1/12)

- 
- Video games
 - Cartoons
 - Visual effects
 - Animated films
 - CAD/CAM
 - Simulation
 - Medical imaging
 - Information visualization

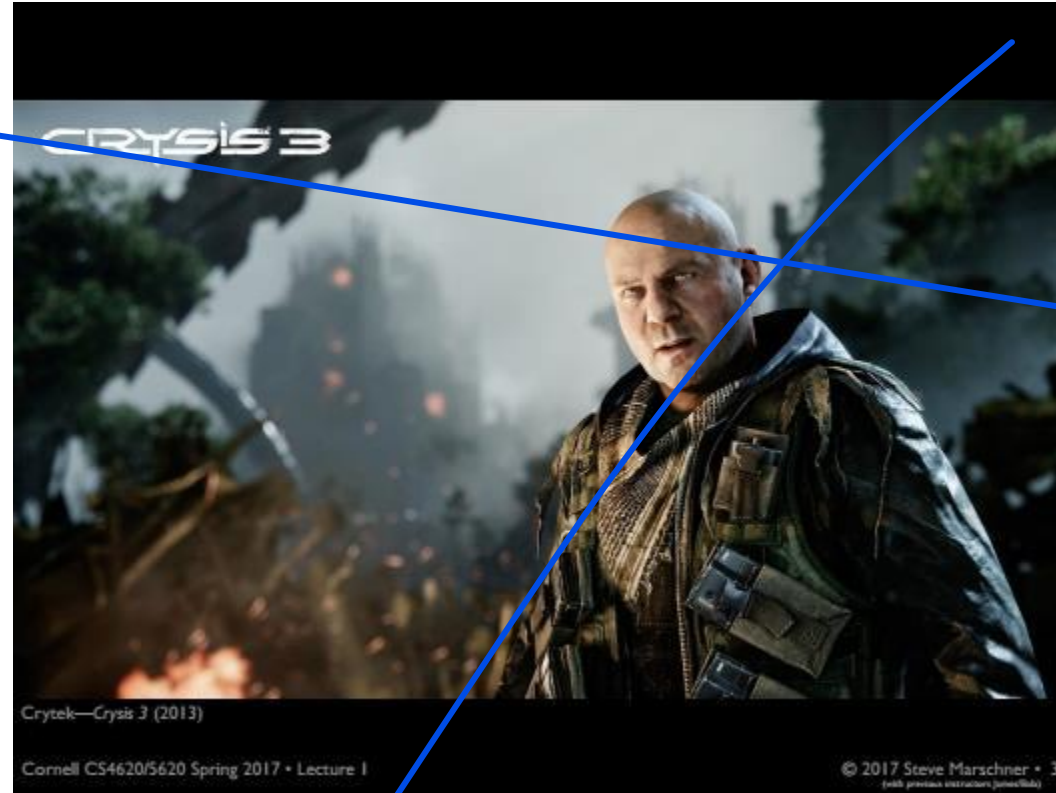
Major Applications (2/12)



Games (2D)

Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | <http://www.cs.cornell.edu/courses/cs4620/2019fa/>

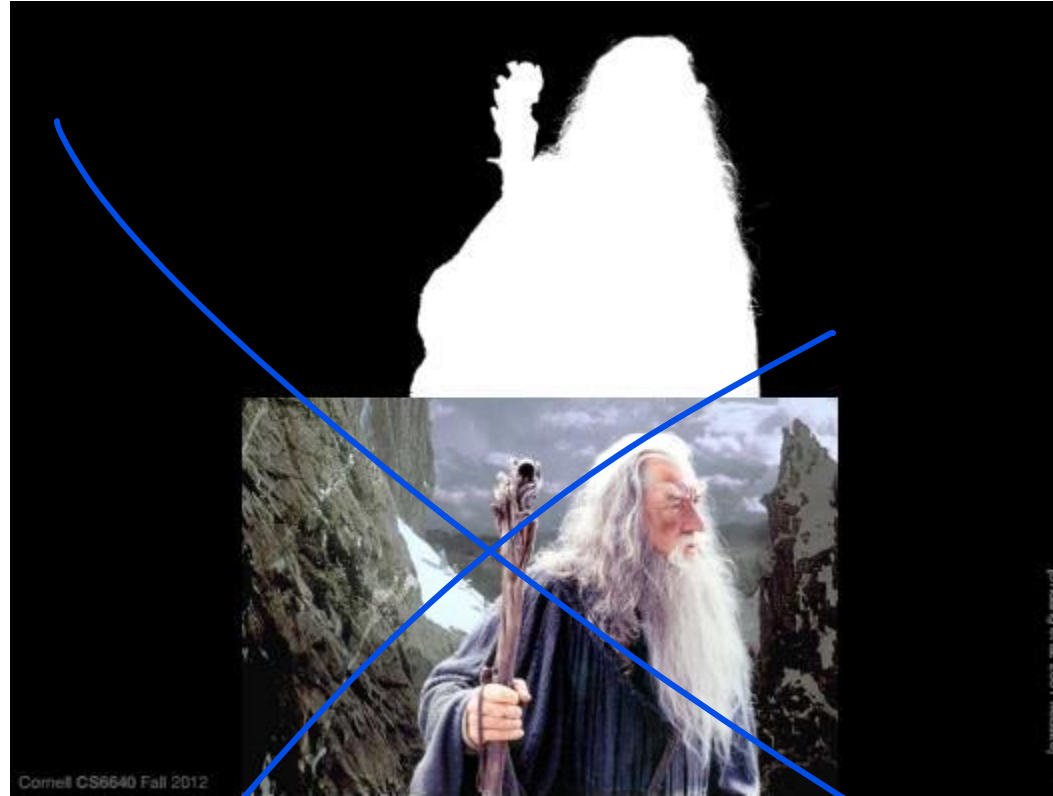
Major Applications (3/12)



Games (3D)

Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | <http://www.cs.cornell.edu/courses/cs4620/2019fa/>

Major Applications (4/12)



Movies (VFX)

Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | <http://www.cs.cornell.edu/courses/cs4620/2019fa/>

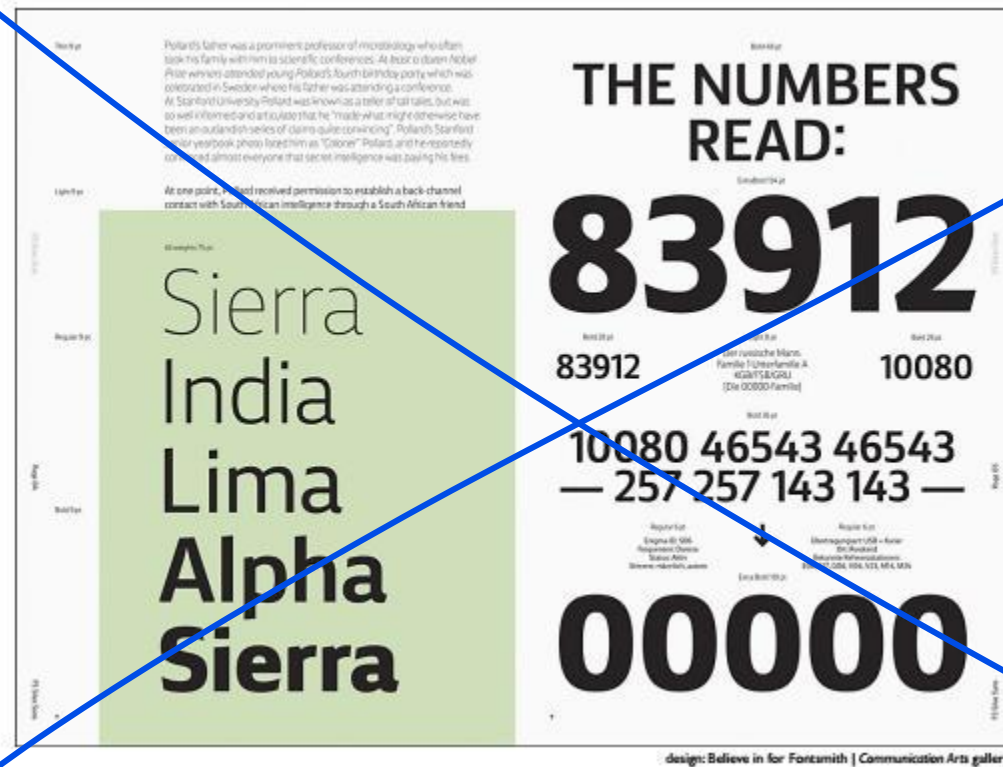
Major Applications (5/12)



Movies (Animated)

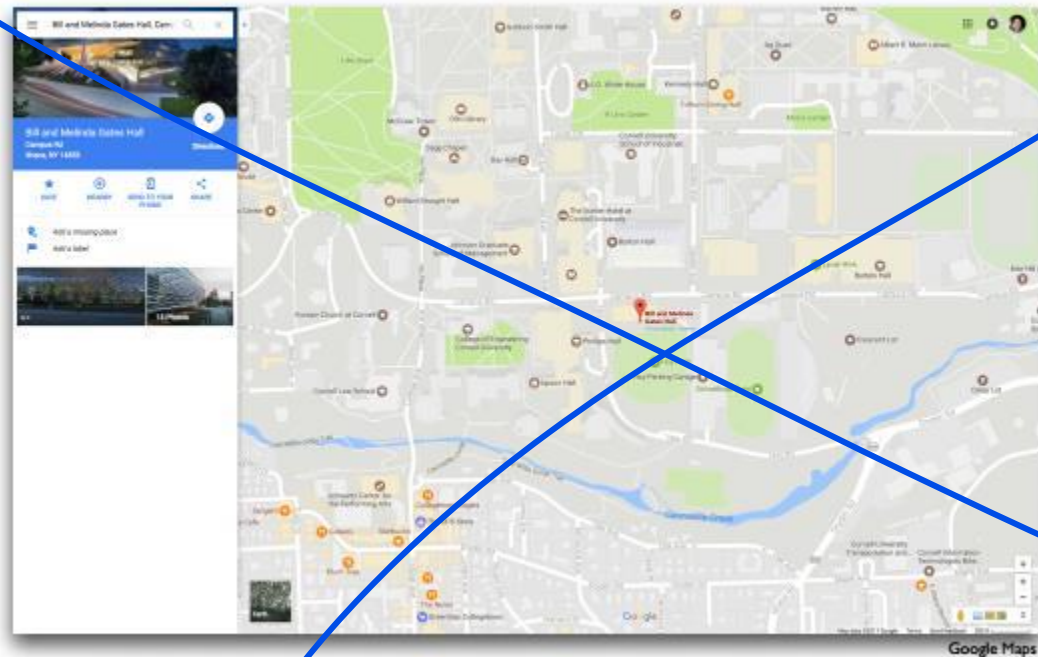
Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | <http://www.cs.cornell.edu/courses/cs4620/2019fa/>

Major Applications (6/12)



Fonts

Major Applications (7/12)



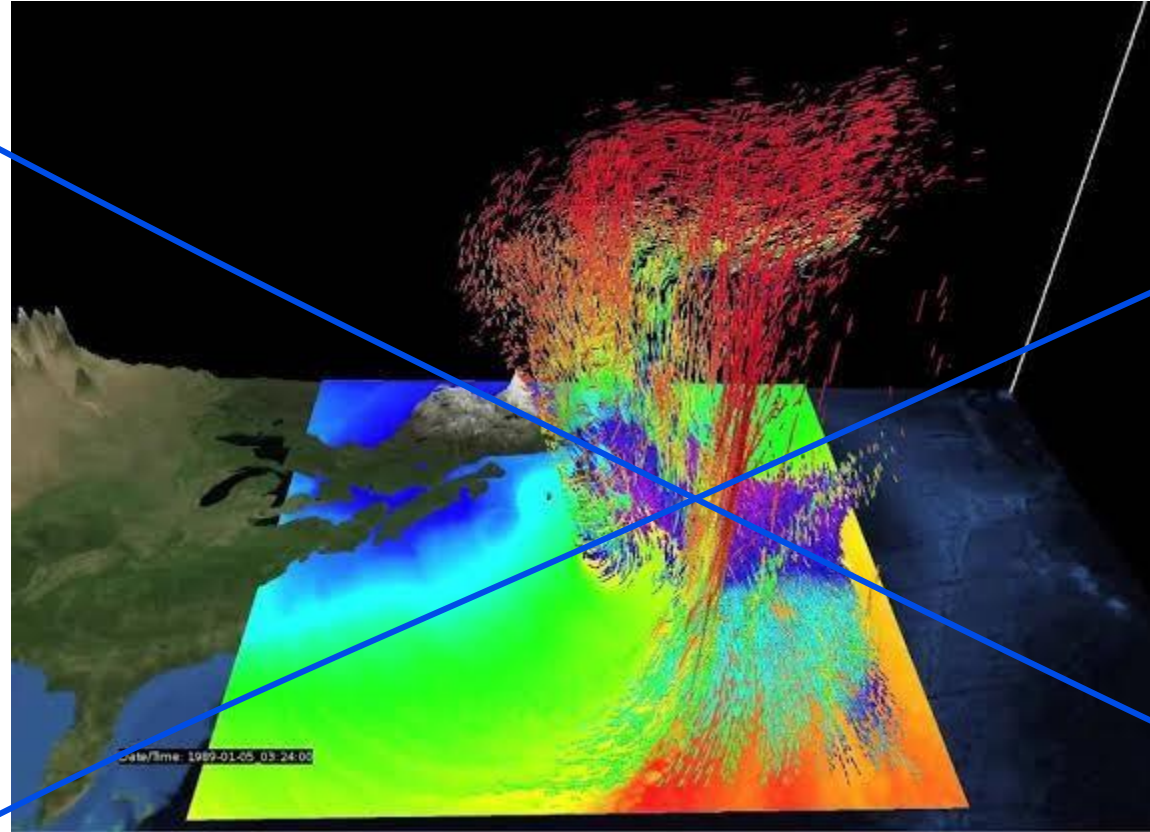
Cornell CS4620/5620 Spring 2017 • Lecture 1

© 2017 Steve Marschner • 19
(with previous instructors' permission)

Google Maps

Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | <http://www.cs.cornell.edu/courses/cs4620/2019fa/>

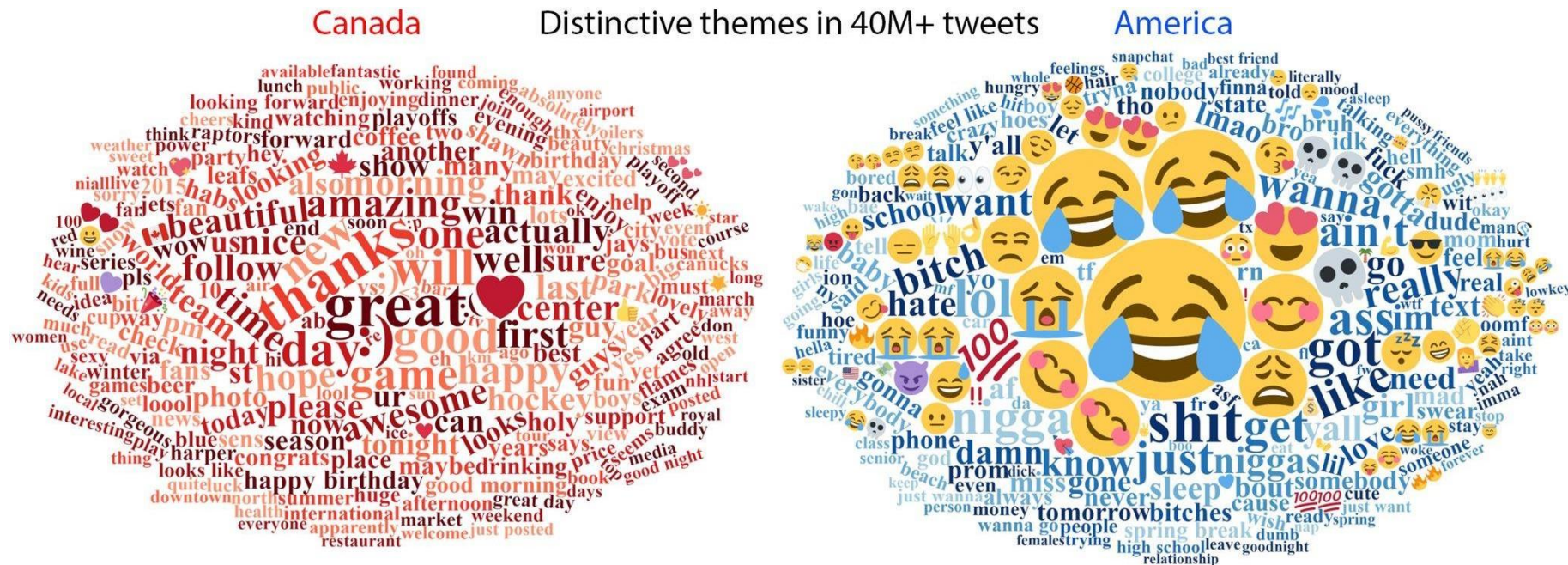
Major Applications (8/12)



Scientific Visualization (SciVis)

Source: <https://youtu.be/eJy5dHMY-S4>

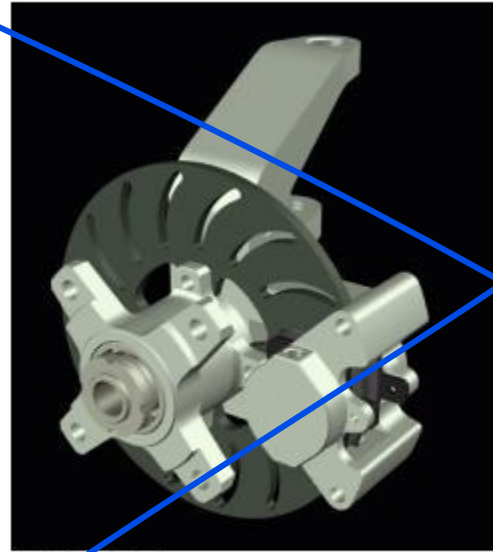
Major Applications (9/12)



Snefjella, Schmidtke, & Kuperman 2018: goo.gl/bqKtqb

Information Visualization (InfoVis)

Major Applications (10/12)



U. of Utah—Alpha 1

Cornell CS4620/5620 Spring 2017 • Lecture 1

© 2017 Steve Marschner • 23
(with previous instructors James/Babe)

CAD (3D modeling)

Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | <http://www.cs.cornell.edu/courses/cs4620/2019fa/>

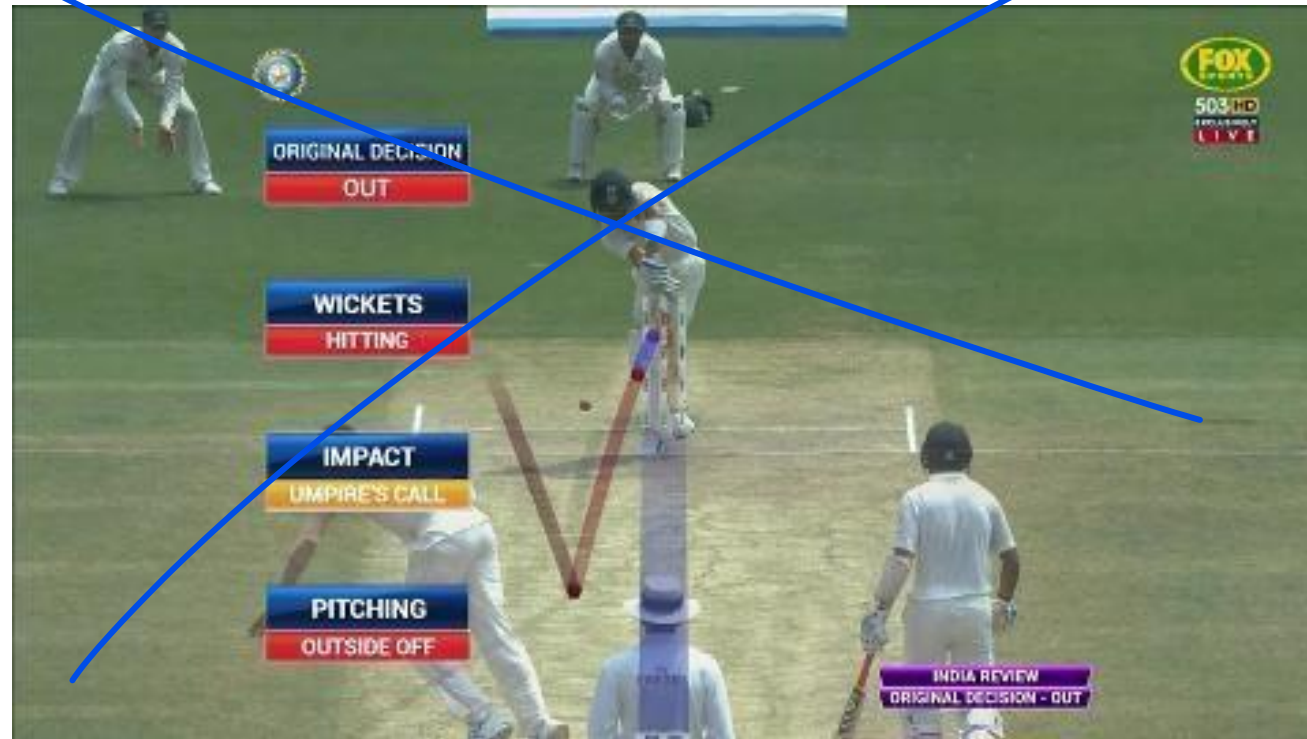
Major Applications (11/12)



Simulation

Source: <https://www.aircharterservice.com/about-us/news-features/blog/are-vr-flight-simulators-the-future-of-pilot-training>

Major Applications (12/12)



Simulation

Source: <https://www.financialexpress.com/sports/what-is-drs-all-the-the-rules-number-of-chances-and-components-explained/578996/>

Graphics API (1/2)

- A **graphics API** is a set of functions that perform basic operations such as –
 - drawing images and 3D surfaces into windows on 2D screen.

Graphics API (2/2)

✓ Every *graphics program* needs to be able to use two related APIs

Application programming interface

– **Graphics API** for visual output.

- Ex:

- i.e. command for drawing lines, circles etc.

– **User-interface API** to get input from the user.

- Ex:

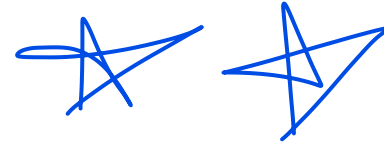
- Window

- Receiving mouse and keyboard input

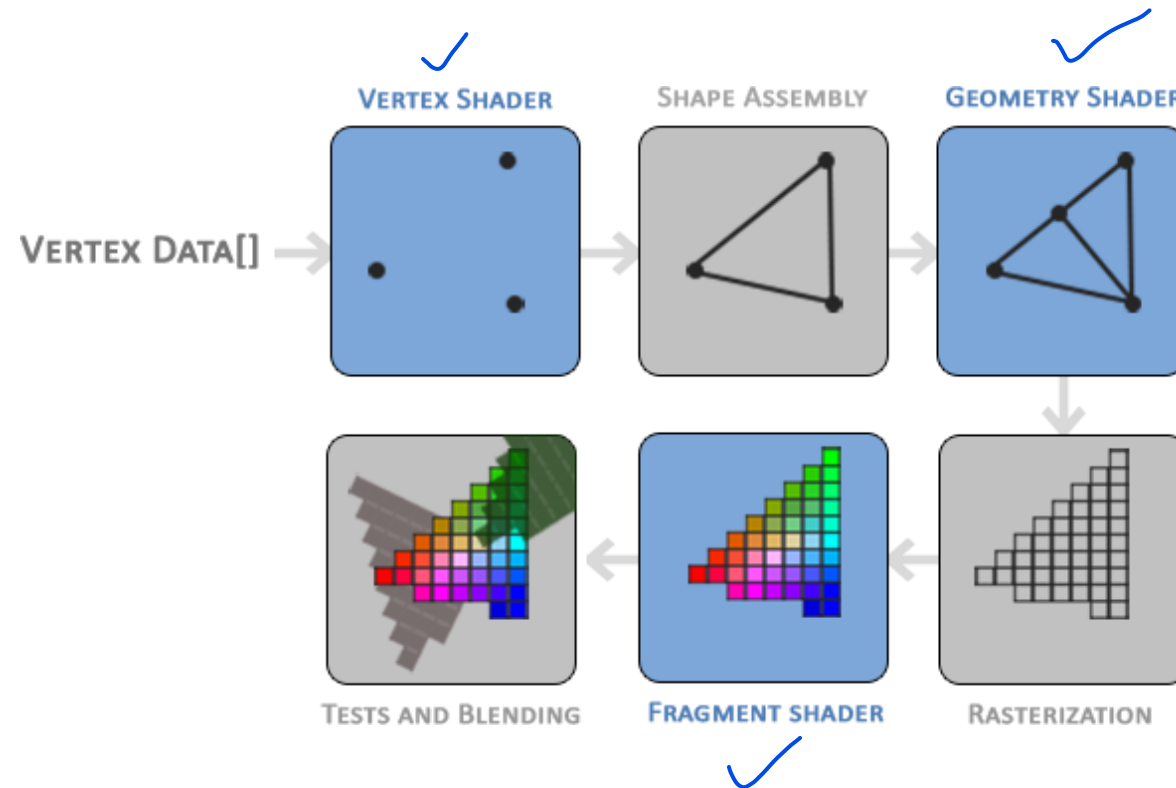
Graphics Pipeline (1/5)

- Special software/hardware subsystem that maps the **3D vertex** locations to **2D screen**.
- From modeling to rendering.
 - Shade the triangles –
 - Realistic
 - Proper back-to-front order.

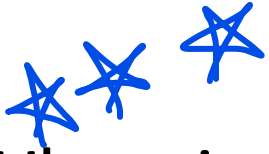
Graphics Pipeline (2/5)



- Pipeline (example):



Graphics Pipeline (3/5)

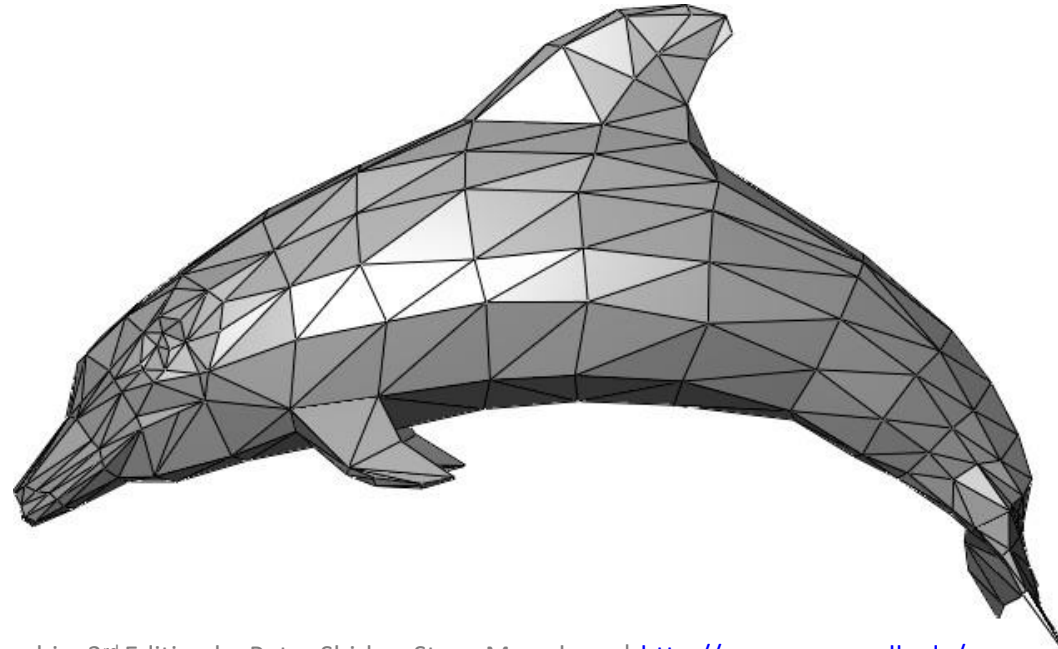


- Why triangles?
 - It is the **simplest** universal surface element
 - it is the **convex hull of three points**.
 - A line or a point are even simpler, but do not create surfaces.
 - it isn't possible to use only a finite number of them without having cracks.

Graphics Pipeline (4/5)

- Mesh:

(A polygon mesh is a collection of vertices, edges and faces that defines the shape of a polyhedral object.
– Ex. *Quad* mesh, Triangle mesh.

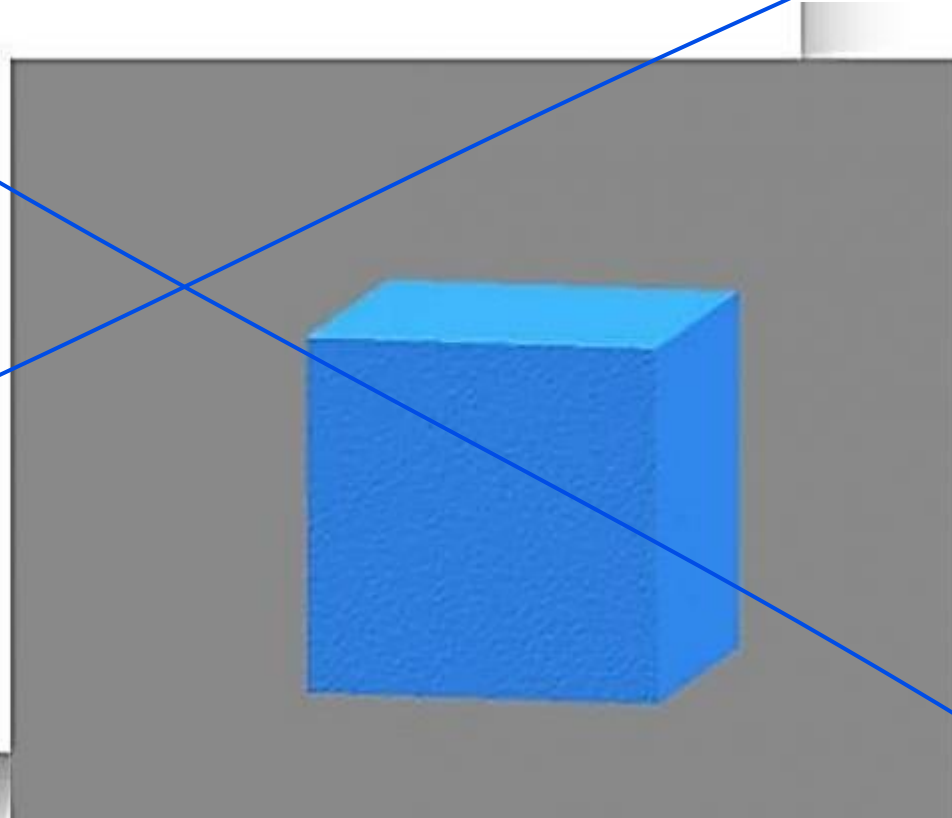


Graphics Pipeline (5/5)

.obj file

```
v 1.000000 1.000000 0.000000
v 1.000000 0.000000 0.000000
v 0.000000 0.000000 0.000000
v 1.000000 1.000000 1.000000
v 0.000000 0.000000 1.000000
v 0.999999 -0.000000 1.000000
v 0.000000 1.000000 1.000000
v 0.000000 1.000000 0.000000
```

```
f 1 2 3
f 1 3 8
f 4 7 5
f 4 5 6
f 1 4 6
f 1 6 2
f 2 6 5
f 2 5 3
f 3 5 7
f 3 7 8
f 4 1 8
f 4 8 7
```



Source: <https://www.sculpteo.com/en/glossary/obj-file-3d-printing-file-format/>

LoD (1/2)

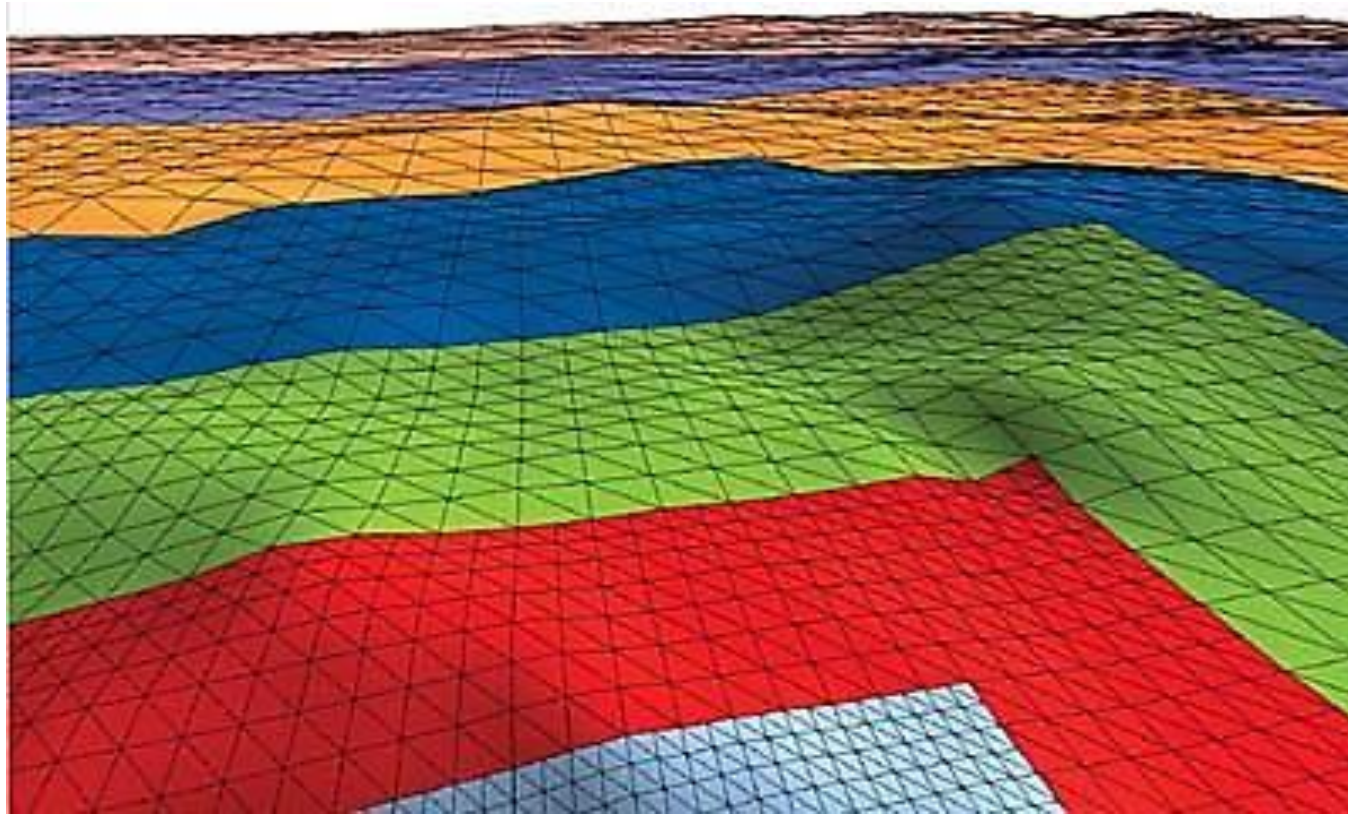
- Rendering speed $\leftrightarrow$ number of triangles being drawn
 - More triangles: more storage.

[0, 0, 3], [0, 3, 3], [0, 3, 0], [0, 0, 0],
[3, 0, 3], [3, 3, 3], [3, 3, 0], [3, 0, 0]

- It is worthwhile to minimize the number of triangles used to represent a model.
 - ✓ Ex: if the model is viewed in the distance, fewer triangles needed (*level of detail or LoD*).

LoD (2/2)

- Example of *LoD*:



Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | <http://www.cs.cornell.edu/courses/cs4620/2019fa/>
Source: <https://developer.nvidia.com/gpugems/gpugems2/part-i-geometric-complexity/chapter-2-terrain-rendering-using-gpu-based-geometry>

Additional Reading

- 1.7: Designing and Coding Graphics Programs

Thank You